



Biology HL/SL A1 Water	Biology HL/SL A1 Water	Biology HL/SL A1 Water
Why does water have an unusually high boiling point compared with other molecules of similar size?	Distinguish the role of cohesion and adhesion of water in xylem transport.	Which property of water allows large bodies of water to buffer temperature fluctuations for the organisms living in them? A High viscosity B High specific heat capacity C Low density as a solid D High surface tension
IB-BIOLOGY-A1-1 CONCEPT	IB-BIOLOGY-A1-2 CONCEPT	IB-BIOLOGY-A1-3 MULTIPLE CHOICE
Biology HL/SL A1 Water	Biology HL/SL A2 Nucleic acids	Biology HL/SL A2 Nucleic acids
Calculate the energy needed to heat 250 g of water from 15°C to 37°C. Specific heat capacity of water = 4.18 J g ⁻¹ °C ⁻¹ .	What structural features allow DNA to store genetic information stably and accurately across generations?	What are the distinct roles of mRNA, tRNA and rRNA in gene expression?
IB-BIOLOGY-A1-4 PRACTICE	IB-BIOLOGY-A2-1 CONCEPT	IB-BIOLOGY-A2-2 CONCEPT
Biology HL/SL A2 Nucleic acids	Biology HL/SL A2 Nucleic acids	Biology HL/SL A3 Cell structure
Which of the following bases is a pyrimidine? A Adenine B Guanine C Cytosine D Both A and B	A DNA strand used as a template reads 5'-CGTATG-3'. Give the sequence of the mRNA transcribed from it.	What is the endosymbiotic theory and what evidence supports it?
IB-BIOLOGY-A2-3 MULTIPLE CHOICE	IB-BIOLOGY-A2-4 PRACTICE	IB-BIOLOGY-A3-1 CONCEPT
Biology HL/SL A3 Cell structure	Biology HL/SL A3 Cell structure	Biology HL/SL A3 Cell structure
Why can a light microscope not resolve individual ribosomes?	Which feature is common to both prokaryotic and eukaryotic cells? A A membrane-bound nucleus B A plasma membrane C Mitochondria D Membrane-bound organelles	An image of a cell measures 25 mm across, and the actual cell is 0.025 mm wide. Calculate the magnification of the image.
IB-BIOLOGY-A3-2 CONCEPT	IB-BIOLOGY-A3-3 MULTIPLE CHOICE	IB-BIOLOGY-A3-4 PRACTICE
Biology HL/SL A4 Diversity of organisms	Biology HL/SL A4 Diversity of organisms	Biology HL/SL A4 Diversity of organisms
What defines a biological species?	Why is rRNA gene sequence data particularly useful for building phylogenies across all domains of life?	In a cladogram, a 'clade' refers to: A Only the living members of a taxonomic group B A common ancestor and all of its descendants C Organisms that share a habitat but not ancestry D Any two organisms with similar appearance
IB-BIOLOGY-A4-1 CONCEPT	IB-BIOLOGY-A4-2 CONCEPT	IB-BIOLOGY-A4-3 MULTIPLE CHOICE
Biology HL/SL A4 Diversity of organisms	Biology HL/SL B1 Carbohydrates and lipids	Biology HL/SL B1 Carbohydrates and lipids
Species A shows 99% DNA sequence similarity to a reference species, while Species B shows 82% similarity. Which is more closely related to the reference species, and why?	What type of reaction joins two monosaccharides together, and what small molecule is produced as a by-product?	Why are unsaturated fats typically liquid at room temperature while saturated fats are solid?
IB-BIOLOGY-A4-4 PRACTICE	IB-BIOLOGY-B1-1 CONCEPT	IB-BIOLOGY-B1-2 CONCEPT



Biology HL/SL B1 Carbohydrates and lipids	Biology HL/SL B1 Carbohydrates and lipids	Biology HL/SL B2 Proteins
Glycogen differs from starch (amylopectin) mainly in that glycogen is: A Made from a different sugar monomer B More highly branched C Joined by peptide bonds D Found only in plant cells	Calculate the energy released from oxidising 5 g of lipid (39 kJ g^{-1}) compared with 5 g of carbohydrate (17 kJ g^{-1}).	What type of bond links amino acids together, and what is released when it forms?
IB-BIOLOGY-B1-3 MULTIPLE CHOICE	IB-BIOLOGY-B1-4 PRACTICE	IB-BIOLOGY-B2-1 CONCEPT
Biology HL/SL B2 Proteins	Biology HL/SL B2 Proteins	Biology HL/SL B2 Proteins
Distinguish fibrous and globular proteins, giving an example and function of each.	Quaternary protein structure refers to: A The folding of a single polypeptide chain B The association of two or more polypeptide chains into one functional protein C The linear sequence of amino acids D The formation of peptide bonds during translation	A mutation changes one amino acid in an enzyme's primary structure. Explain how this could affect the enzyme's function.
IB-BIOLOGY-B2-2 CONCEPT	IB-BIOLOGY-B2-3 MULTIPLE CHOICE	IB-BIOLOGY-B2-4 PRACTICE
Biology HL/SL B3 Membranes and membrane transport	Biology HL/SL B3 Membranes and membrane transport	Biology HL/SL B3 Membranes and membrane transport
What does the fluid mosaic model describe?	Distinguish facilitated diffusion from active transport.	Osmosis is best defined as the net movement of: A Solute particles across a membrane B Water across a partially permeable membrane from a region of higher to lower water potential C Any substance down its concentration gradient D Ions through specific channel proteins only
IB-BIOLOGY-B3-1 CONCEPT	IB-BIOLOGY-B3-2 CONCEPT	IB-BIOLOGY-B3-3 MULTIPLE CHOICE
Biology HL/SL B3 Membranes and membrane transport	Biology HL/SL B4 Gas exchange	Biology HL/SL B4 Gas exchange
A red blood cell is placed in distilled water. Explain what happens to the cell and why.	List the features shared by all efficient gas exchange surfaces.	Why is countercurrent flow in fish gills more efficient at extracting oxygen than concurrent flow would be?
IB-BIOLOGY-B3-4 PRACTICE	IB-BIOLOGY-B4-1 CONCEPT	IB-BIOLOGY-B4-2 CONCEPT
Biology HL/SL B4 Gas exchange	Biology HL/SL B4 Gas exchange	Biology HL/SL B5 Organisms exchange with their environment (HL)
During human inhalation, which change occurs? A The diaphragm relaxes and domes upward B The diaphragm contracts and flattens, increasing thoracic volume C Intercostal muscles have no effect on breathing D Thoracic volume decreases	Explain briefly why increasing physical activity increases a person's breathing rate.	Distinguish 'source' and 'sink' in phloem transport.
IB-BIOLOGY-B4-3 MULTIPLE CHOICE	IB-BIOLOGY-B4-4 PRACTICE	IB-BIOLOGY-B5-1 CONCEPT



Biology HL/SL B5 Organisms exchange with their environment (HL) Why does having a double circulatory system benefit mammals compared with a single circulatory system? IB-BIOLOGY-B5-2 CONCEPT	Biology HL/SL B5 Organisms exchange with their environment (HL) Which structural feature helps xylem vessels withstand the tension generated during transpiration pull? A Living cytoplasm inside the vessel B Lignified cell walls that resist collapse under tension C Numerous small perforations in the walls D A thick waxy cuticle covering the vessel IB-BIOLOGY-B5-3 MULTIPLE CHOICE	Biology HL/SL B5 Organisms exchange with their environment (HL) In spring, a deciduous tree mobilises sugars stored in its roots to support new shoot growth, reversing its usual summer transport pattern. Explain this change in terms of source and sink. IB-BIOLOGY-B5-4 PRACTICE
Biology HL/SL C1 Enzymes and metabolism How does an enzyme increase the rate of a reaction without being used up? IB-BIOLOGY-C1-1 CONCEPT	Biology HL/SL C1 Enzymes and metabolism Distinguish competitive and non-competitive inhibition in terms of where they bind and their effect on Vmax. IB-BIOLOGY-C1-2 CONCEPT	Biology HL/SL C1 Enzymes and metabolism Raising the temperature above an enzyme's optimum usually decreases reaction rate because: A Substrate concentration falls B Enzyme molecules move more slowly C Bonds maintaining the enzyme's tertiary structure break, changing the active site shape D The enzyme concentration decreases IB-BIOLOGY-C1-3 MULTIPLE CHOICE
Biology HL/SL C1 Enzymes and metabolism Explain why the rate of an enzyme-catalysed reaction plateaus at high substrate concentrations even though substrate is still in excess. IB-BIOLOGY-C1-4 PRACTICE	Biology HL/SL C2 Cell respiration Why does anaerobic respiration yield far less ATP than aerobic respiration from the same amount of glucose? IB-BIOLOGY-C2-1 CONCEPT	Biology HL/SL C2 Cell respiration What is the role of oxygen in aerobic cell respiration? IB-BIOLOGY-C2-2 CONCEPT
Biology HL/SL C2 Cell respiration Lactic acid fermentation in muscle cells regenerates which molecule that is essential for glycolysis to continue? A ATP B NAD ⁺ C Oxygen D Pyruvate IB-BIOLOGY-C2-3 MULTIPLE CHOICE	Biology HL/SL C2 Cell respiration A yeast culture is moved from aerobic to anaerobic conditions. Explain how ATP yield per glucose molecule changes, and predict what happens to the rate of glucose consumption if energy demand stays the same. IB-BIOLOGY-C2-4 PRACTICE	Biology HL/SL C3 Photosynthesis What are the products of the light-dependent reactions used for in the Calvin cycle? IB-BIOLOGY-C3-1 CONCEPT
Biology HL/SL C3 Photosynthesis Why does the rate of photosynthesis plateau at high light intensity even when CO ₂ is not limiting? IB-BIOLOGY-C3-2 CONCEPT	Biology HL/SL C3 Photosynthesis Which pigment mainly absorbs light in the blue-violet and red regions of the spectrum, reflecting green light? A A carotenoid B Chlorophyll C A xanthophyll D A phycobilin IB-BIOLOGY-C3-3 MULTIPLE CHOICE	Biology HL/SL C3 Photosynthesis Explain briefly why increasing CO ₂ concentration increases the rate of photosynthesis up to a certain point, but not beyond it. IB-BIOLOGY-C3-4 PRACTICE



Biology HL/SL C4 Ecosystems and energy flow	Biology HL/SL C4 Ecosystems and energy flow	Biology HL/SL C4 Ecosystems and energy flow
Distinguish gross primary production (GPP) and net primary production (NPP).	Why do food chains rarely contain more than four or five trophic levels?	Which biological process returns carbon fixed by producers back to the atmosphere as CO ₂ ? A Photosynthesis B Cell respiration C Transpiration D Nitrogen fixation
IB-BIOLOGY-C4-1 CONCEPT	IB-BIOLOGY-C4-2 CONCEPT	IB-BIOLOGY-C4-3 MULTIPLE CHOICE
Biology HL/SL C4 Ecosystems and energy flow	Biology HL/SL C5 Populations and communities (HL)	Biology HL/SL C5 Populations and communities (HL)
Producers in a field fix 6000 kJ m ⁻² yr ⁻¹ of net energy. Assuming a 10% transfer efficiency at each trophic transfer, calculate the energy available to secondary consumers (two transfers away from the producers).	Distinguish density-dependent and density-independent limiting factors on population size.	Why is there typically a time lag between a peak in prey population and a peak in predator population?
IB-BIOLOGY-C4-4 PRACTICE	IB-BIOLOGY-C5-1 CONCEPT	IB-BIOLOGY-C5-2 CONCEPT
Biology HL/SL C5 Populations and communities (HL)	Biology HL/SL C5 Populations and communities (HL)	Biology HL/SL D1 DNA replication
Which scenario is an example of interspecific competition? A Two prides of the same lion species fighting over territory B A wolf population and a bear population both feeding on the same limited berry patch C A lion population preying on a zebra population D A mutualistic relationship between a clownfish and a sea anemone	A population grows exponentially, then plateaus at its carrying capacity K. Name and explain one density-dependent factor that could cause this plateau.	What does it mean that DNA replication is 'semi-conservative'?
IB-BIOLOGY-C5-3 MULTIPLE CHOICE	IB-BIOLOGY-C5-4 PRACTICE	IB-BIOLOGY-D1-1 CONCEPT
Biology HL/SL D1 DNA replication	Biology HL/SL D1 DNA replication	Biology HL/SL D1 DNA replication
Why can't DNA polymerase begin synthesising a new DNA strand without a primer?	DNA polymerase synthesises a new DNA strand by adding nucleotides: A In the 3' to 5' direction only B In the 5' to 3' direction, extending the 3' end of the growing strand C Randomly along the template D In both directions simultaneously without needing a primer	Explain briefly why the lagging strand of DNA is synthesised as a series of short fragments rather than continuously.
IB-BIOLOGY-D1-2 CONCEPT	IB-BIOLOGY-D1-3 MULTIPLE CHOICE	IB-BIOLOGY-D1-4 PRACTICE
Biology HL/SL D2 Protein synthesis	Biology HL/SL D2 Protein synthesis	Biology HL/SL D2 Protein synthesis
Why is the genetic code described as 'degenerate'?	Distinguish exons and introns in eukaryotic gene expression.	What happens when a ribosome reaches a stop codon on the mRNA? A Another amino acid is added to the chain B The completed polypeptide is released and the ribosome dissociates from the mRNA C Transcription of a new gene begins D An intron is removed from the mRNA
IB-BIOLOGY-D2-1 CONCEPT	IB-BIOLOGY-D2-2 CONCEPT	IB-BIOLOGY-D2-3 MULTIPLE CHOICE



Biology HL/SL D2 Protein synthesis	Biology HL/SL D3 Cell division	Biology HL/SL D3 Cell division
A base substitution changes an mRNA codon from AAA (lysine) to AGA (arginine). Name this type of mutation and predict its likely effect on the protein. IB-BIOLOGY-D2-4PRACTICE	How does the chromosome number of daughter cells compare between mitosis and meiosis? IB-BIOLOGY-D3-1CONCEPT	Name two processes during meiosis that generate genetic variation, and briefly explain each. IB-BIOLOGY-D3-2CONCEPT
Biology HL/SL D3 Cell division	Biology HL/SL D3 Cell division	Biology HL/SL D4 Genetics and inheritance
During which stage of mitosis do sister chromatids separate and move toward opposite poles of the cell? A Prophase B Metaphase C Anaphase D Telophase IB-BIOLOGY-D3-3MULTIPLE CHOICE	Explain briefly why uncontrolled cell division can lead to tumour formation. IB-BIOLOGY-D3-4PRACTICE	Distinguish genotype and phenotype. IB-BIOLOGY-D4-1CONCEPT
Biology HL/SL D4 Genetics and inheritance	Biology HL/SL D4 Genetics and inheritance	Biology HL/SL D4 Genetics and inheritance
Why do X-linked recessive conditions appear more frequently in males than in females? IB-BIOLOGY-D4-2CONCEPT	Two heterozygous pea plants (Tt x Tt) for a trait where T (tall) is dominant to t (dwarf) are crossed. What proportion of offspring is expected to be dwarf? A 1/4 B 1/2 C 3/4 D 0 IB-BIOLOGY-D4-3MULTIPLE CHOICE	A colour-blind (X-linked recessive) man and a woman who is homozygous for normal vision have children. Predict the genotypes and phenotypes of their daughters and sons. IB-BIOLOGY-D4-4PRACTICE
Biology HL/SL D5 Natural selection and evolution	Biology HL/SL D5 Natural selection and evolution	Biology HL/SL D5 Natural selection and evolution
What four conditions are necessary for natural selection to occur in a population? IB-BIOLOGY-D5-1CONCEPT	Why can't natural selection act on a non-heritable trait, such as a scar gained during an animal's lifetime? IB-BIOLOGY-D5-2CONCEPT	Which of the following is an example of homologous structures providing evidence for evolution from a common ancestor? A An insect wing and a bird wing, which share a function but have very different underlying structures B A human arm and a bat wing, which have a similar underlying bone arrangement but different functions C Fish gills and human lungs, which have completely different structures and functions D None of the above are homologous structures IB-BIOLOGY-D5-3MULTIPLE CHOICE
Biology HL/SL D5 Natural selection and evolution	Biology HL/SL D6 Gene expression and homeostasis (HL)	Biology HL/SL D6 Gene expression and homeostasis (HL)
Explain briefly why overuse of antibiotics tends to increase the frequency of antibiotic-resistant bacteria in a population. IB-BIOLOGY-D5-4PRACTICE	How can a liver cell and a muscle cell, with identical DNA, have very different structures and functions? IB-BIOLOGY-D6-1CONCEPT	What is the key difference between a mutation and an epigenetic change in terms of their effect on DNA? IB-BIOLOGY-D6-2CONCEPT



Biology HL/SL

D6 Gene expression and homeostasis (HL)

A steroid hormone typically regulates gene expression in a target cell by:

- A Binding a cell-surface receptor and triggering a second-messenger cascade
- B Diffusing through the plasma membrane and binding an intracellular receptor that then acts as a transcription factor
- C Binding directly to mRNA molecules in the cytoplasm
- D Directly inhibiting DNA polymerase during replication

IB-BIOLOGY-D6-3

MULTIPLE CHOICE

Biology HL/SL

D6 Gene expression and homeostasis (HL)

Explain briefly why switching genes off is just as important as switching genes on during cell differentiation.

IB-BIOLOGY-D6-4

PRACTICE



Biology HL/SL A1 Water	Biology HL/SL A1 Water	Biology HL/SL A1 Water
B. A high specific heat capacity means water absorbs or releases large amounts of heat with only small temperature changes, stabilising the thermal environment. IB-BIOLOGY-A1-3 ANSWER	Cohesion is the attraction between water molecules (via hydrogen bonds) that holds the water column together under tension; adhesion is the attraction between water molecules and the xylem vessel walls, which helps resist the pull of gravity. IB-BIOLOGY-A1-2 ANSWER	Its molecules form hydrogen bonds with each other; a large amount of energy is needed to break these numerous intermolecular bonds before the molecules can separate into gas. IB-BIOLOGY-A1-1 ANSWER
Biology HL/SL A2 Nucleic acids	Biology HL/SL A2 Nucleic acids	Biology HL/SL A1 Water
mRNA carries the genetic code (as codons) from DNA in the nucleus to the ribosome; tRNA carries a specific amino acid and uses its anticodon to recognise the matching mRNA codon; rRNA forms the structural and catalytic core of the ribosome. IB-BIOLOGY-A2-2 ANSWER	The sugar-phosphate backbone gives a stable double helix, and specific hydrogen-bonded complementary base pairing (A-T, C-G) between the two antiparallel strands protects the coding sequence and allows it to be copied accurately during replication. IB-BIOLOGY-A2-1 ANSWER	$q = m \times c \times \Delta T = 250 \times 4.18 \times (37 - 15) = 250 \times 4.18 \times 22 = 22\,990 \text{ J} \approx 23.0 \text{ kJ}.$ IB-BIOLOGY-A1-4 ANSWER
Biology HL/SL A3 Cell structure	Biology HL/SL A2 Nucleic acids	Biology HL/SL A2 Nucleic acids
It proposes mitochondria and chloroplasts evolved from free-living prokaryotes engulfed by an ancestral eukaryotic cell. Evidence: double membranes, their own circular DNA, their own 70S ribosomes, and the ability to divide independently of the host cell. IB-BIOLOGY-A3-1 ANSWER	Pairing complementary bases and reading in the correct antiparallel direction gives an mRNA of 5'-CAUACG-3' (complementary and antiparallel to the template, with U replacing T). IB-BIOLOGY-A2-4 ANSWER	C. Cytosine (along with thymine and uracil) is a pyrimidine, a single-ring nitrogenous base; adenine and guanine are purines (double-ring). IB-BIOLOGY-A2-3 ANSWER
Biology HL/SL A3 Cell structure	Biology HL/SL A3 Cell structure	Biology HL/SL A3 Cell structure
Magnification = image size ÷ actual size = $25 \text{ mm} \div 0.025 \text{ mm} = 1000\times.$ IB-BIOLOGY-A3-4 ANSWER	B. All cells, prokaryotic and eukaryotic, are enclosed by a plasma membrane that regulates exchange with the environment. IB-BIOLOGY-A3-3 ANSWER	Resolution is limited by the wavelength of the radiation used; visible light has a wavelength (~400–700 nm) too long to distinguish objects as small as ribosomes (~20–30 nm), so they appear as a single blur or are invisible. IB-BIOLOGY-A3-2 ANSWER
Biology HL/SL A4 Diversity of organisms	Biology HL/SL A4 Diversity of organisms	Biology HL/SL A4 Diversity of organisms
B. A clade is a monophyletic group: an ancestral lineage together with every species descended from it. IB-BIOLOGY-A4-3 ANSWER	rRNA genes are found in every living organism and are highly conserved because of their essential role in the ribosome, so differences accumulate slowly and steadily, allowing evolutionary relatedness to be compared even between very distantly related organisms. IB-BIOLOGY-A4-2 ANSWER	A group of organisms that can interbreed under natural conditions and produce viable, fertile offspring. IB-BIOLOGY-A4-1 ANSWER



Biology HL/SL B1 Carbohydrates and lipids	Biology HL/SL B1 Carbohydrates and lipids	Biology HL/SL A4 Diversity of organisms
Unsaturated fatty acids contain C=C double bonds that create kinks in the chain, preventing molecules from packing closely together, which lowers the melting point; saturated fatty acids have straight chains that pack tightly, raising the melting point.	A condensation reaction forms a glycosidic bond between the monosaccharides and releases a molecule of water.	Species A is more closely related, because a higher percentage sequence similarity indicates fewer accumulated mutations since divergence, implying a more recent common ancestor.
IB-BIOLOGY-B1-2 ANSWER	IB-BIOLOGY-B1-1 ANSWER	IB-BIOLOGY-A4-4 ANSWER
Biology HL/SL B2 Proteins	Biology HL/SL B1 Carbohydrates and lipids	Biology HL/SL B1 Carbohydrates and lipids
A peptide bond, formed by a condensation reaction that releases a molecule of water.	Lipid: $5 \times 39 = 195$ kJ. Carbohydrate: $5 \times 17 = 85$ kJ. The lipid releases 110 kJ more energy than the same mass of <u>carbohydrate</u> .	B. Glycogen has a higher degree of branching than amylopectin, allowing faster mobilisation of glucose to meet animal metabolic <u>demands</u> .
IB-BIOLOGY-B2-1 ANSWER	IB-BIOLOGY-B1-4 ANSWER	IB-BIOLOGY-B1-3 ANSWER
Biology HL/SL B2 Proteins	Biology HL/SL B2 Proteins	Biology HL/SL B2 Proteins
Changing one amino acid can alter the pattern of hydrogen, ionic or hydrophobic interactions, changing the tertiary structure and therefore the shape of the active site; if the active site shape is no longer complementary to the substrate, the enzyme's catalytic activity is reduced or lost (though some substitutions have little or no effect).	B. Quaternary structure describes how multiple polypeptide subunits combine, e.g. the four subunits of haemoglobin.	Fibrous proteins (e.g. collagen) form long, insoluble strands with a structural role; globular proteins (e.g. enzymes, haemoglobin) fold into compact, soluble 3D shapes suited to functional roles such as catalysis or transport.
IB-BIOLOGY-B2-4 ANSWER	IB-BIOLOGY-B2-3 ANSWER	IB-BIOLOGY-B2-2 ANSWER
Biology HL/SL B3 Membranes and membrane transport	Biology HL/SL B3 Membranes and membrane transport	Biology HL/SL B3 Membranes and membrane transport
B. Osmosis specifically refers to the passive net movement of water molecules across a partially permeable membrane, from high to low water potential.	Facilitated diffusion is passive movement of a substance down its concentration gradient through a channel or carrier protein, requiring no ATP; active transport moves a substance against its concentration gradient using a pump protein powered by ATP.	It describes the plasma membrane as a bilayer of phospholipids in which proteins are embedded or attached; the phospholipids and many proteins can move laterally, giving the membrane fluidity ('fluid') and a varied protein composition ('mosaic').
IB-BIOLOGY-B3-3 ANSWER	IB-BIOLOGY-B3-2 ANSWER	IB-BIOLOGY-B3-1 ANSWER
Biology HL/SL B4 Gas exchange	Biology HL/SL B4 Gas exchange	Biology HL/SL B3 Membranes and membrane transport
Countercurrent flow keeps blood's oxygen concentration below that of the adjacent water along the entire gill lamella, maintaining a diffusion gradient throughout; in concurrent flow, the gradient disappears once blood and water oxygen concentrations equalise partway <u>along, limiting extraction</u> .	A large surface area, a short/thin diffusion distance, a moist surface for gases to dissolve in, and a good blood supply or ventilation that maintains a steep concentration gradient.	The cell swells and may burst (lyse), because distilled water has a higher water potential than the cell's cytoplasm, so water enters the cell by osmosis down the water potential gradient; unlike plant cells, red blood cells lack a rigid cell wall to resist the resulting pressure.
IB-BIOLOGY-B4-2 ANSWER	IB-BIOLOGY-B4-1 ANSWER	IB-BIOLOGY-B3-4 ANSWER



Biology HL/SL B5 Organisms exchange with their environment (HL)	Biology HL/SL B4 Gas exchange	Biology HL/SL B4 Gas exchange
A source is a region that releases sugar into the phloem (e.g. a photosynthesising leaf, or a storage organ mobilising its reserves); a sink is a region that removes sugar from the phloem to use or store it (e.g. a growing root or fruit). Transport always runs from source to sink, though which organ plays which role can change over time.	Increased respiration produces more CO₂, lowering blood pH; chemoreceptors detect this and increase ventilation rate/depth to remove CO₂ faster and take in more O₂, restoring homeostasis.	B. Diaphragm contraction flattens it, increasing thoracic (and therefore lung) volume, which lowers internal air pressure and draws air in.
IB-BIOLOGY-B5-1 ANSWER	IB-BIOLOGY-B4-4 ANSWER	IB-BIOLOGY-B4-3 ANSWER
Biology HL/SL B5 Organisms exchange with their environment (HL)	Biology HL/SL B5 Organisms exchange with their environment (HL)	Biology HL/SL B5 Organisms exchange with their environment (HL)
In spring the root becomes the source (releasing stored sugar into the phloem) and the growing shoot/buds become the sink (using the sugar for growth), so sap flow direction in the phloem reverses compared with summer, when photosynthesising leaves are the source and the root is the sink.	B. Lignin strengthens and stiffens xylem vessel walls, preventing them from collapsing inward under the negative pressure (tension) generated by transpiration.	Double circulation separates the pulmonary circuit (heart to lungs) from the systemic circuit (heart to body); blood is re-pressurised by the heart after passing through the low-pressure lung capillaries, maintaining high pressure in the systemic circuit, enabling faster delivery of oxygen to support a higher metabolic rate.
IB-BIOLOGY-B5-4 ANSWER	IB-BIOLOGY-B5-3 ANSWER	IB-BIOLOGY-B5-2 ANSWER
Biology HL/SL C1 Enzymes and metabolism	Biology HL/SL C1 Enzymes and metabolism	Biology HL/SL C1 Enzymes and metabolism
C. Excess kinetic energy above the optimum disrupts hydrogen and ionic bonds holding the tertiary structure, denaturing the active site and reducing activity.	Competitive inhibitors resemble the substrate and bind the active site, competing with substrate; their effect can be overcome by increasing substrate concentration, so V_{max} is unchanged. Non-competitive inhibitors bind elsewhere (an allosteric site), changing the enzyme's shape; this lowers V_{max} and cannot be overcome by adding more substrate.	It binds its substrate(s) at the active site and stabilises the transition state, lowering the activation energy needed for the reaction, providing an alternative lower-energy pathway; the unchanged enzyme is then released to catalyse further reactions.
IB-BIOLOGY-C1-3 ANSWER	IB-BIOLOGY-C1-2 ANSWER	IB-BIOLOGY-C1-1 ANSWER
Biology HL/SL C2 Cell respiration	Biology HL/SL C2 Cell respiration	Biology HL/SL C1 Enzymes and metabolism
Oxygen acts as the final electron acceptor in the electron transport chain; it combines with electrons and H⁺ ions to form water, allowing electron flow (and the proton gradient driving ATP synthase) to continue, so ATP production via oxidative phosphorylation can proceed.	Without oxygen as the final electron acceptor, the electron transport chain cannot operate, so ATP is generated only by the substrate-level phosphorylation of glycolysis (net 2 ATP); the Krebs cycle and oxidative phosphorylation, which produce the majority of ATP (around 30 total) in aerobic respiration, cannot proceed.	At high substrate concentrations, all available enzyme active sites are continuously occupied (the enzyme is saturated); enzyme concentration, not substrate concentration, becomes the limiting factor, so rate cannot increase further until more enzyme is added.
IB-BIOLOGY-C2-2 ANSWER	IB-BIOLOGY-C2-1 ANSWER	IB-BIOLOGY-C1-4 ANSWER



Biology HL/SL C3 Photosynthesis	Biology HL/SL C2 Cell respiration	Biology HL/SL C2 Cell respiration
ATP supplies energy and NADPH supplies reducing power (hydrogen/electrons) to convert CO ₂ fixed onto RuBP (forming a 3-carbon compound) into G3P; some G3P regenerates RuBP and some is used to synthesise glucose and other organic molecules.	ATP yield per glucose drops sharply, from roughly 30 (aerobic) to 2 (anaerobic, alcoholic fermentation), because oxidative phosphorylation cannot occur without oxygen; to meet the same energy demand, the yeast must consume glucose at a much faster rate under anaerobic conditions.	B. Reducing pyruvate to lactate regenerates NAD ⁺ from NADH, allowing glycolysis to continue producing ATP even without oxygen.
IB-BIOLOGY-C3-1 ANSWER	IB-BIOLOGY-C2-4 ANSWER	IB-BIOLOGY-C2-3 ANSWER
Biology HL/SL C3 Photosynthesis	Biology HL/SL C3 Photosynthesis	Biology HL/SL C3 Photosynthesis
CO ₂ is the substrate fixed by rubisco in the Calvin cycle, so more CO ₂ increases the rate of carbon fixation up to the point where the enzymes become saturated; beyond that point another factor, such as light intensity or temperature, becomes limiting instead.	B. Chlorophyll strongly absorbs blue-violet and red light and reflects green light, giving most photosynthetic tissue its green colour.	Another factor, such as temperature or the maximum capacity of Calvin cycle enzymes (e.g. rubisco), becomes limiting; once these enzymes are working at their maximum rate they cannot process additional ATP/NADPH any faster, so the overall rate levels off.
IB-BIOLOGY-C3-4 ANSWER	IB-BIOLOGY-C3-3 ANSWER	IB-BIOLOGY-C3-2 ANSWER
Biology HL/SL C4 Ecosystems and energy flow	Biology HL/SL C4 Ecosystems and energy flow	Biology HL/SL C4 Ecosystems and energy flow
B. Cell respiration in producers, consumers and decomposers oxidises organic molecules, releasing CO ₂ back into the atmosphere as part of the carbon cycle.	Only about 10% of the energy at one trophic level is typically transferred to the next (the rest is lost as heat via respiration, or in undigested/unconsumed matter), so the energy available shrinks rapidly with each transfer until too little remains to support a further trophic level.	GPP is the total amount of energy/biomass fixed by producers through photosynthesis; NPP is GPP minus the energy producers use in their own respiration, representing the energy actually available to consumers and for producer growth.
IB-BIOLOGY-C4-3 ANSWER	IB-BIOLOGY-C4-2 ANSWER	IB-BIOLOGY-C4-1 ANSWER
Biology HL/SL C5 Populations and communities (HL)	Biology HL/SL C5 Populations and communities (HL)	Biology HL/SL C4 Ecosystems and energy flow
It takes time for increased prey (food) availability to be converted into greater predator survival and successful reproduction (e.g. through gestation and offspring development); so predator numbers only rise substantially, and later peak, some time after the prey population has already peaked.	Density-dependent factors (e.g. competition, predation, disease) have a greater effect on population growth as population density increases, driving logistic growth toward carrying capacity K; density-independent factors (e.g. a storm, fire, extreme cold) reduce population size regardless of its density.	After the first transfer to primary consumers: $6000 \times 0.10 = 600 \text{ kJ m}^{-2} \text{ yr}^{-1}$. After the second transfer to secondary consumers: $600 \times 0.10 = 60 \text{ kJ m}^{-2} \text{ yr}^{-1}$.
IB-BIOLOGY-C5-2 ANSWER	IB-BIOLOGY-C5-1 ANSWER	IB-BIOLOGY-C4-4 ANSWER
Biology HL/SL D1 DNA replication	Biology HL/SL C5 Populations and communities (HL)	Biology HL/SL C5 Populations and communities (HL)
Each of the two new DNA molecules produced by replication contains one strand from the original (parental) molecule and one newly synthesised strand, so half of each original molecule is conserved in every daughter molecule.	Example: food limitation — as population size increases, per capita food availability decreases, intensifying competition, which lowers birth rate and/or raises death rate until the two balance, stabilising the population at K.	B. Interspecific competition occurs between individuals of different species competing for the same limited resource, here berries shared by wolves and bears.
IB-BIOLOGY-D1-1 ANSWER	IB-BIOLOGY-C5-4 ANSWER	IB-BIOLOGY-C5-3 ANSWER



Biology HL/SL D1 DNA replication	Biology HL/SL D1 DNA replication	Biology HL/SL D1 DNA replication
Because DNA polymerase synthesises only 5'→3' and the lagging strand's template runs in the opposite direction to the replication fork's movement, synthesis on this strand must happen in short bursts back toward the fork as the helix unwinds, producing discontinuous Okazaki fragments that are later joined by DNA ligase. IB-BIOLOGY-D1-4ANSWER	B. New nucleotides are always added to the free 3' end of the growing strand, so synthesis proceeds 5' to 3'. IB-BIOLOGY-D1-3ANSWER	DNA polymerase can only add nucleotides onto an existing free 3'-OH end; it cannot join the first two nucleotides of a new strand together, so a short RNA/DNA primer must first be laid down to provide that starting 3'-OH group. IB-BIOLOGY-D1-2ANSWER
Biology HL/SL D2 Protein synthesis	Biology HL/SL D2 Protein synthesis	Biology HL/SL D2 Protein synthesis
B. A stop codon does not code for an amino acid; it signals release factors to terminate translation, releasing the finished polypeptide. IB-BIOLOGY-D2-3ANSWER	Exons are the coding sequences of a gene that remain in the mature mRNA and are translated; introns are non-coding sequences within the pre-mRNA that are removed by splicing before translation. IB-BIOLOGY-D2-2ANSWER	Most amino acids are specified by more than one codon (synonymous codons), often differing only in the third ('wobble') base, meaning a single-base change does not always change the amino acid produced. IB-BIOLOGY-D2-1ANSWER
Biology HL/SL D3 Cell division	Biology HL/SL D3 Cell division	Biology HL/SL D2 Protein synthesis
Crossing over: homologous chromosomes exchange segments of DNA during prophase I, creating new allele combinations on each chromosome. Independent assortment: homologous chromosome pairs (and later, sister chromatids) are distributed to daughter cells in random combinations, so different gametes receive different mixes of maternal and paternal chromosomes. IB-BIOLOGY-D3-2ANSWER	Mitosis maintains the diploid chromosome number ($2n \rightarrow 2n$), producing two genetically identical daughter cells; meiosis halves the chromosome number ($2n \rightarrow n$), producing four genetically variable haploid cells (gametes). IB-BIOLOGY-D3-1ANSWER	This is a missense mutation. One amino acid in the polypeptide changes from lysine to arginine; the effect on protein function depends on how chemically similar the two amino acids are and where in the protein the change occurs — it may have little effect, or could disrupt folding/function if the position or property change is significant. IB-BIOLOGY-D2-4ANSWER
Biology HL/SL D4 Genetics and inheritance	Biology HL/SL D3 Cell division	Biology HL/SL D3 Cell division
Genotype is the specific combination of alleles an organism has for a gene; phenotype is the observable physical, physiological or biochemical characteristic that results from the genotype, often also influenced by the environment. IB-BIOLOGY-D4-1ANSWER	Mutations in genes that normally regulate the cell cycle (e.g. checkpoint genes) allow cells to bypass normal controls and divide repeatedly without stopping, producing an abnormal, growing mass of cells (a tumour). IB-BIOLOGY-D3-4ANSWER	C. In anaphase, the centromere splits and spindle fibres pull the separated sister chromatids toward opposite poles. IB-BIOLOGY-D3-3ANSWER

**Biology HL/SL**

D4 Genetics and inheritance

All daughters inherit the father's X^b and the mother's X^B , so they are heterozygous carriers ($X^B X^b$) with normal vision phenotypically. All sons inherit the mother's X^B and the father's Y , so they are $X^B Y$, with normal vision (they do not inherit their father's X chromosome).

IB-BIOLOGY-D4-4

ANSWER

Biology HL/SL

D4 Genetics and inheritance

A. Only the homozygous recessive genotype tt (1 of the 4 equally likely combinations) produces the dwarf phenotype.

IB-BIOLOGY-D4-3

ANSWER

Biology HL/SL

D4 Genetics and inheritance

Males are hemizygous for X-linked genes (XY), so a single recessive allele on their one X chromosome is enough to cause the phenotype; females (XX) must be homozygous recessive (inherit the recessive allele on both X chromosomes) to be affected, since one dominant allele is sufficient to mask the recessive phenotype.

IB-BIOLOGY-D4-2

ANSWER

Biology HL/SL

D5 Natural selection and evolution

B. Homologous structures share an underlying anatomical plan inherited from a common ancestor, even where the function has diverged, as with the human arm and bat wing.

IB-BIOLOGY-D5-3

ANSWER

Biology HL/SL

D5 Natural selection and evolution

Natural selection changes allele frequencies across generations, which requires that a trait be encoded in DNA and passed to offspring; a non-heritable trait is not transmitted to the next generation, so differential survival based on it cannot change the genetic composition of future populations.

IB-BIOLOGY-D5-2

ANSWER

Biology HL/SL

D5 Natural selection and evolution

There must be heritable variation in a trait; the trait must affect an individual's survival or reproductive success; the environment must exert a selective pressure favouring some variants over others; and this must lead to differential reproductive success that changes allele frequencies in the next generation.

IB-BIOLOGY-D5-1

ANSWER

Biology HL/SL

D6 Gene expression and homeostasis (HL)

A mutation changes the actual base sequence of the DNA; an epigenetic change (such as methylation or histone modification) alters gene expression/chromatin accessibility without changing the underlying DNA sequence, and is potentially reversible.

IB-BIOLOGY-D6-2

ANSWER

Biology HL/SL

D6 Gene expression and homeostasis (HL)

Differential gene expression: different combinations of genes are transcribed and translated in each cell type due to different active transcription factors and different epigenetic modifications (e.g. methylation patterns), even though the underlying genome is identical.

IB-BIOLOGY-D6-1

ANSWER

Biology HL/SL

D5 Natural selection and evolution

The antibiotic acts as a strong selective pressure: bacteria carrying a (randomly arising) resistance allele survive and reproduce, while susceptible bacteria are killed, so the proportion of the resistance allele increases in the surviving population with each generation of exposure — an example of natural selection.

IB-BIOLOGY-D5-4

ANSWER

Biology HL/SL

D6 Gene expression and homeostasis (HL)

Differentiating cells must stop expressing genes appropriate to other cell types or to an undifferentiated (stem) state, restricting expression to only the genes needed for their specific specialised function; without gene silencing, a cell could not establish or maintain a distinct, specialised identity.

IB-BIOLOGY-D6-4

ANSWER

Biology HL/SL

D6 Gene expression and homeostasis (HL)

B. Being lipid-soluble, steroid hormones cross the plasma membrane and bind intracellular receptors; the hormone-receptor complex then acts in the nucleus as a transcription factor, regulating specific gene expression.

IB-BIOLOGY-D6-3

ANSWER